

USERSTUDY

Study Goal

Evaluate the **user-friendliness** (ease of use, intuitiveness, and satisfaction) of the *Fortunately–Unfortunately Tree Security Modeler* prototype.

Participants

- **Target group:** cybersecurity students, software engineers, or people with moderate technical literacy.
 - **Sample size:** 5–10 participants for testing;
 - **Inclusion criteria:**
 - Basic understanding of threat modeling or security risk reasoning.
 - No prior exposure to this tool (to avoid bias).
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Study Design Overview

Phase	Duration	Description
1. Introduction	5 min	Explain purpose (testing usability, not participant). Obtain consent.
2. Training / Demo	5 min	Give short demo of core functions.
3. Task Session	25–30 min	Participants perform realistic tasks using the app. Measure effectiveness, efficiency, satisfaction.
4. Questionnaire	10 min	Fill usability/satisfaction survey (SUS or custom Likert-scale).
5. Interview / Reflection	10 min	Collect open feedback on experience, design, and suggestions.

Tasks (core usability features)

Each task maps directly to your implemented features and helps evaluate *discoverability, efficiency, and understandability*.

Task	Measured Features	Success Criteria	Observed Metrics
T1. Create a new tree	<code>handleAddTree()</code>	Participant creates a root node successfully.	Completion rate, time, confusion.
T2. Expand and collapse nodes	<code>GoChevronDown/Up</code> buttons	Participant successfully expands/collapses subtrees.	Number of wrong clicks, time.
T3. Rename a node	On-click rename prompt	Participant renames a node correctly.	Discoverability, task success.
T4. Add child nodes	<code>RiAddBoxLine</code>	Participant adds one or more children, alternating levels (fortunate/unfortunate)	Accuracy, ease of discovery.
T5. Delete a node	<code>GoTrash</code>	Participant deletes a specific child node.	Errors, confirmation understanding.
T6. Search and highlight nodes	<code>SearchBar</code>	Participant finds nodes by keyword, navigates through results with Enter.	Accuracy, satisfaction.
T7. View parent node info (tooltip)	Tooltip logic	Participant hovers to view parent info and interprets it correctly.	Success, clarity rating.
T10. Use sidebar tabs	Side menu (<code>Weaknesses</code> , <code>Assumptions</code> , etc.)	Participant navigates node lists and interprets colors/states.	Task time, comprehension.
T11. Reset the tree	<code>handleReset()</code>	Participant resets tree after confirmation prompt.	Observed caution, confirmation comprehension.

Quantitative Metrics

- **Task completion rate (%)**
- **Average task completion time (s)**
- **Error rate** (e.g., incorrect action or confusion pause >5s)
- **System Usability Scale (SUS)**
 - Standard questionnaire (1–5 Likert scale)

- Produces overall usability score (0–100).
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Qualitative Questions (Post-test Interview)

General Perception

- What did you like most about the interface?
- What confused or frustrated you?
- Was it clear what “fortunate” and “unfortunate” meant in the context?
- How did you feel about the tooltip and sidebar features?
- If you could change one thing to make it easier to use, what would it be?

Visual & Interaction Design

- How did you experience the color coding (red/green)?
- Was the sidebar organization intuitive?
- Did the search feel helpful and responsive?

Learning Curve

- How quickly did you feel confident using the software?
 - Would you need additional explanations or tutorials?
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TASKS

1. Create a New Tree

Goal: Test discoverability of **Add Tree** button

Task instruction:

“Start from an empty screen and create a new tree.”

Expected actions: Click the **Add Tree** button → root node appears.

Follow-up questions:

- How easy was it to find where to start?
- Was it clear what “Add Tree” would do?
- What would you expect to happen next?

2. Add Child Nodes

Goal: Test use of `Add Node` (`RiAddBoxLine`) and color/state awareness

Task instruction:

"Add two new child nodes under the root node — one fortunate and one unfortunate."

Follow-up questions:

- Was it obvious how to add new nodes?
 - How did you know the difference between fortunate and unfortunate nodes?
 - Did the colors (green/red) make sense to you?
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3. Expand and Collapse Nodes

Goal: Evaluate discoverability of expand/collapse icons (`GoChevronDown` / `GoChevronUp`)

Task instruction:

"Collapse all nodes and then expand the tree again."

"Collapse part ... if branch"

Follow-up questions:

- Did you recognize what the arrow icons meant immediately?
 - Would you expect another way to expand/collapse nodes?
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4. Rename a Node

Goal: Evaluate in-place editing or rename feature

Task instruction:

"Rename one of your nodes to something else that fits the scenario."

Follow-up questions:

- Was it clear how to rename a node?
 - Did you expect to click on it or find a rename button?
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5. Delete a Node

Goal: Evaluate deletion safety and clarity (`GoTrash`)

Task instruction:

"Delete one of the child nodes you created."

Follow-up questions:

- Did you notice the delete icon easily?
 - Were you sure it would delete that node (not the whole tree)?
 - Did you expect a confirmation message?
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6. Search and Highlight

Goal: Test `SearchBar` with highlighting and navigation

Task instruction:

"Use the search bar at the top to find the node ... and add a child to it. Is it an unfortunate or a fortunate you add?"

"Press Enter a few times to move between results."

Follow-up questions:

- Did the highlighting make it clear which node was selected?
 - Would you expect to filter or just highlight results?
 - Did the search feel fast and intuitive?
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7. View Tooltip Information

Goal: Evaluate tooltip hover interaction

Task instruction:

"Hover over a node to view information about its parent."

Follow-up questions:

- Was it clear that hovering would show more info?
 - Was the tooltip content useful or confusing?
 - Did the tooltip disappear at the right time (not too fast or slow)?
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8. Explore the Sidebar Tabs

Goal: Test navigation of right-side menu (**Weaknesses** , **Assumptions** , **Strengths** , **Opportunities**)

Task instruction:

"Open the sidebar and explore the different tabs (Weaknesses, Assumptions, Strengths, Opportunities)."

"Try to interpret what each section shows."

"What assumptions does the tree make"

"What are use-full branches that fixes security problems of the software"

"What are some possible problems/weaknesses that are not fixed yet"

"Imagine you want to see all the assumptions in your model. Which icon would you click?"

"Now, which icon would you click if you wanted to check a fortunate nodes for possible unfortunates?"

Follow-up questions:

- Were the tab names meaningful to you?
 - Did the colors or layout help you understand what's listed?
 - Did switching tabs feel responsive?
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9. Reset the Tree

Goal: Check understanding of **Reset Tree** button and confirmation dialogue

Task instruction:

"Now reset your tree using the 'Reset Tree' button."

Follow-up questions:

- Did you hesitate before clicking reset?
 - Was the confirmation message clear enough?
 - Did you expect the tree to reset completely?
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10. Interpret the Icon Buttons in the Side Panel

Goal: Evaluate whether users can correctly understand and use the **icon buttons** in the right-hand side panel — without relying on text labels.

Task instruction:

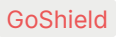
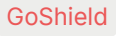
"Look at the icons when hovering over a node.

Without clicking anything yet, try to tell what you think each icon represents or does."

"Change the status of a node so it fits the sidebar button discription. Add an assumption that needs further verification to the verified assumptions tab in the side bar"

(Ask them to say aloud or note their interpretation for each icon)

Example Icons & Intended Meanings:

Icon (example)	Intended Meaning	Expected User Description
	Strengths	"Represents security strengths or positive factors/fixed problems."
 red	Weaknesses	"Shows weaknesses or vulnerabilities."
 orange of a fortunately node	Investigation needed	"States the nodes that still need extra investigation"
 orange of a unfortunately node	Assumptions to verify	"Has solutions but need further verification"

Follow-up questions:

- Without hovering or clicking, could you tell what each icon meant?
- Which icons were immediately clear, and which were confusing?
- Did the icons match your expectations once you clicked them?
- Did the color or position help you understand their meaning?
- Would you prefer text labels, tooltips, or explanations next to the icons?
- Did you notice any icons that looked similar or ambiguous?

Optional Quantitative Rating Questions (1–5 scale)

Question	Scale
How easy was it to understand what each button does?	1 (very hard) – 5 (very easy)

Question	Scale
How confident did you feel while creating and editing nodes?	1–5
How visually clear was the tree structure?	1–5
How helpful were the colors and icons?	1–5
How intuitive was the search and highlight function?	1–5
How organized did the sidebar feel?	1–5
How likely would you be to use this tool again?	1 (not at all) – 5 (definitely)
The meaning of each icon was clear without explanation.	1 (strongly disagree) – 5 (strongly agree)
The icons fit well with their function.	1–5
I could easily navigate using the icons alone.	1–5
I would prefer text labels next to icons.	1 (no) – 5 (definitely yes)

Optional Add-ons

If you want more depth:

- **Screen recording** (observe live reasoning).
- **NASA-TLX** workload scale to measure perceived effort.

Example Research Questions

1. How intuitive is the process of building and managing a Fortunately–Unfortunately tree for security modeling?
2. Do visual cues (colors, icons, tooltips) enhance user comprehension of node relationships?
3. How efficient and satisfying do users find navigation (search, expand/collapse, and sidebar)?
4. What usability improvements can be identified from user behavior and feedback?
5. What part of the interface did you find most intuitive?
6. What part was confusing or took time to figure out?

7. Did you find any button labels unclear or misleading?
 8. Did you notice any feature by accident?
 9. If you could change one thing to make it easier to use, what would it be?
 10. How do you feel about the way the tool visualizes "fortunate" vs "unfortunate" nodes?
 11. Did the interactions (hover, click, expand) behave the way you expected?
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Data Analysis Plan

Data Type	Analysis Method
Task performance	Calculate success rate, mean completion time per task. Compare across participants.
SUS scores	Compute average SUS; interpret according to usability benchmarks (e.g., >68 = "good").
Open feedback	Thematic analysis (group comments by recurring themes: clarity, aesthetics, confusion, etc.).
Observations	Log misclicks, hesitation, and verbalized difficulties.